FSK 116 – Semester 1
Mathematics and Other Essentials

Priorities

- Know how YOUR calculator works and always have YOUR calculator with you.
- Always have a pencil (and an eraser) at hand when doing Physics.

Greek Alphabet

Αα	Alpha	Ηη	Eta	Νν	Nu	Ττ	Tau
Ββ	Beta	Θθ	Theta	Ξξ	Xi	Υυ	Upsilon
Γγ	Gamma	Ιι	Iota	Οο	Omicron	Φφ	Phi
Δ δ	Delta	Κκ	Kappa	Ππ	Pi	Χχ	Chi
Εε	Epsilon	Λλ	Lambda	Ρρ	Rho	Ψψ	Psi
Ζζ	Zeta	Μμ	Mu	Σ σ	Sigma	Ω ω	Omega

Greek Alphabet cont.

α	Alpha Not **ɑ**	η	Eta Not n	ν	Nu Not v	υ	Upsilon Not u
β	Beta Not B	κ	Kappa Not k, K	ρ	Rho Not p	χ	Chi Not X
ε	Epsilon Not E	μ	Mu Not u	τ ᴛ	Tau Not T,t	ω	Omega Not w

Symbols and Constants

$=$	Equal to	$\neq$	Not equal to	$\sim$	Same order of magnitude	$\approx$	Approx. equal to
$\equiv$	Identical	$\propto$	Proportional to	$<$	Less than	$>$	Greater Than
$<<$	Much less than	$>>$	Much greater than	$\geq$	Greater than or equal to	$\leq$	Less than or equal to
∞	Because	$\therefore$	Therefore	$\Rightarrow$	Implies, from this follows	$\rightarrow$	Tends to
Δx, δx	Change in x	$\lvert a \rvert$	Absolute value of a	n!	= n(n-1)(n-2)… (3)(2)(1)	$\sum_{i=1}^{n} x_i$	Sum of all the values of x
π	=3.141592…	e	= 2.71828183				

Prefixes – SI Units

tera	T	10^{12}	centi	c	10^{-2}
giga	G	10^{9}	milli	m	10^{-3}
mega	M	10^{6}	micro	µ	10^{-6}
kilo	k	10^{3}	nano	n	10^{-9}
hecto	h	10^{2}	pico	p	10^{-12}
deka	da	10^{1}	femto	f	10^{-15}
deci	d	10^{-1}	atto	a	10^{-18}

Examples

- Write the following using the correct prefixes and 2 or 3 significant figures:
  - 1 234 567 mm
  - 46 351 259 378 bytes
  - 0.000 003 89 m
  - 0.000 000 000 000 054 s
  - 7 891 N
  - 654 321 J

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Exponents

- *When n is real and positive:*
  - $a^n = a \times a \times a \times a....\ \text{n times}$
  - $a^{-n} = {}^{1}\!/_{a^n} = \dfrac{1}{a \times a \times a \times a....\ \text{n times}}$
  - $a^{1/n} = \sqrt[n]{a} \qquad a^{1/3} = \sqrt[3]{a}$
  - $a^p \times a^q = a^{p+q}$
  - ${}^{a^p}\!/_{a^q} = a^p \div a^q = a^{p-q}$
- *If a>0 and a^n=b, then b>0*
- *a is the base, n is the exponent and b is the n^{th} power of a.*
- *When using base e:* $e = 2.71828183$

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Examples

- Determine the following in exponent notation:
  - $10^2 \times 10^3 =$
  - $\left(3\times10^2\right)\left(2\times10^0\right)=$
  - $\dfrac{a^5}{a^3}=$
  - $\dfrac{10^0}{10^{-3}}=$
  - $\dfrac{8\times10^2}{2\times10^{-6}}=$

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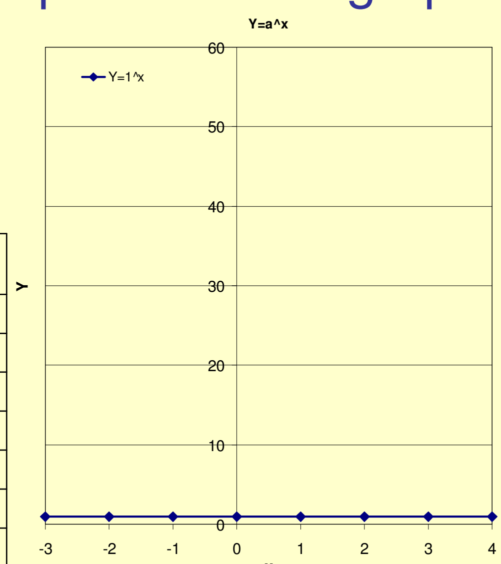
Example – Exponents and graphs

- $y = a^x$
- For a=1
- For a>1, eg. a = 2
- For a = ½
- For a = e

x	$y = a^x$ (a = 1)	$y = a^x$ (a = 2)	$y = a^x$ (a = ½)	$y = a^x$ ($a = e$)
-3	1			
-2	1			
-1	1			
0	1			
1	1			
2	1			
3	1			
4	1			



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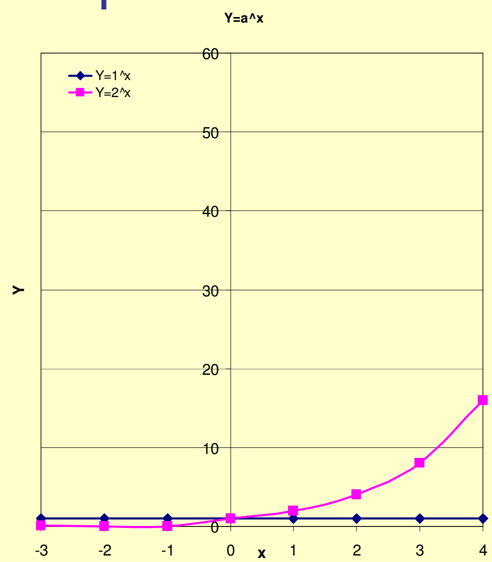
Example cont1

- $y = a^x$
- For a=1
- For a>1, eg. a = 2
- For a = ½
- For a = e

x	$y = a^x$ (a = 1)	$y = a^x$ (a = 2)	$y = a^x$ (a = ½)	$y = a^x$ (a = e)
-3	1	1/8		
-2	1	¼		
-1	1	½		
0	1	1		
1	1	2		
2	1	4		
3	1	8		
4	1	16		



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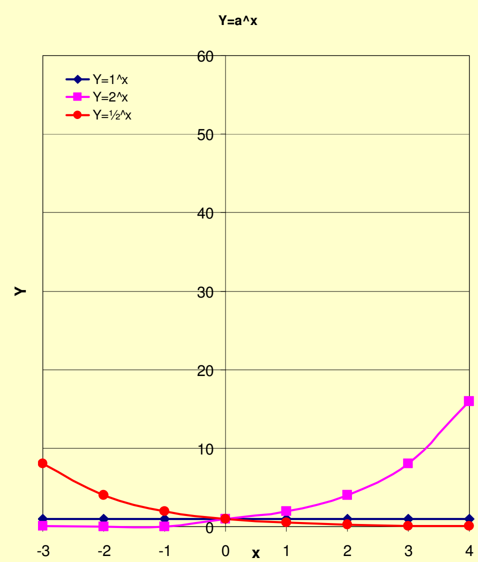
Example cont2

- $y = a^x$
- For a=1
- For a>1, eg. a = 2
- For a = ½
- For a = e

x	$y = a^x$ (a = 1)	$y = a^x$ (a = 2)	$y = a^x$ (a = ½)	$y = a^x$ (a = e)
-3	1	1/8	8	
-2	1	¼	4	
-1	1	½	2	
0	1	1	1	
1	1	2	½	
2	1	4	¼	
3	1	8	1/8	
4	1	16	1/16	



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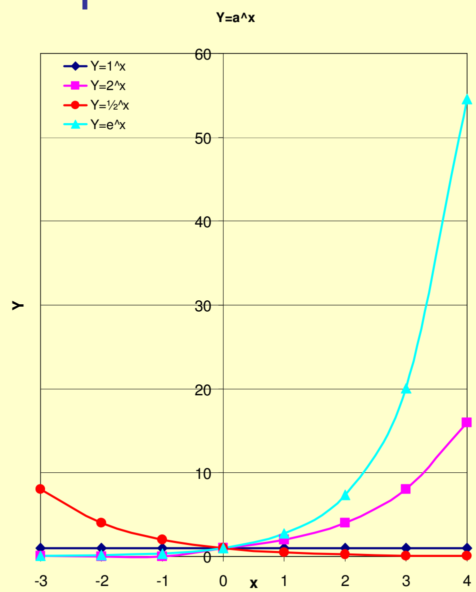
Example cont3

- $y = a^x$
- For a=1
- For a>1, eg. a = 2
- For a = ½
- For a = e

x	$y = a^x$ ($a = 1$)	$y = a^x$ ($a = 2$)	$y = a^x$ ($a = ½$)	$y = a^x$ ($a = e$)
-3	1	1/8	8	0.05
-2	1	¼	4	0.13
-1	1	½	2	0.37
0	1	1	1	1
1	1	2	½	2.72
2	1	4	¼	7.39
3	1	8	1/8	20.09
4	1	16	1/16	54.60



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Logarithms & Natural Logs

- *If* $b = a^n$, *then* $n = \log_a b$
- *Also* $a = b^{1/n}$, *then* $1/n = \log_b a$

$$\Rightarrow \log_b a = 1/\log_a b$$

- *If a=10 then base is usually left out - if* $r = \log s$ *, then* $s = 10^r$
- *Base* e *– natural logarithm –* $e = 2.71828183$
  so if $\ln s \equiv \log_e s = r$ *(pronounced "lin s"),*
  then $s = e^r$

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Rules for Logs

- If *a, P, Q* are positive, then
  - $\log_a P^n = n \log_a P$
  - $\log_a \sqrt[n]{P} = \frac{1}{n} \log_a P$
  - $\log_a 1 = 0$
  - $\log_a \left(\frac{1}{P}\right) = -\log_a P$
  - $\log_a (P \times Q) = \log_a P + \log_a Q$
  - $\log_a \left(\frac{P}{Q}\right) = \log_a P - \log_a Q$
  - $\log_a P = (\log_b P) \times (\log_a b)$

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Examples

- Determine the following
  - $10^3 =$
  - $\text{Log } 1000 =$
  - $10^2 =$
  - $\text{Log } 100 =$
  - $10^{1.5432} =$
  - $\text{Log } 34.93 =$
  - $10^0 =$
  - $\text{Log } 1 =$
  - $10^{-1.5432} =$
  - $\text{Log } 0.02863 =$

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Examples

- Determine *n* in the following:
  - $n = \log 123.4 =$
  - $n = \log 12.34 =$
  - $n = \log 1.234 =$
  - $n = \log\left(123.4\right)^2 =$
  - $n = \left(\log 123.4\right)^2 =$
  - $n = \ln 123.4 =$
  - $n = \ln\left(1/e^2\right) =$

Application of Logs: Earthquakes

- Richter Scale
  - Not accurate for large quakes

$$M_L = \log A - \log A_0 = \log \frac{A}{A_0}$$

  - *A* is maximum excursion of the Wood-Anderson seismograph.
  - A_0 depends only on the epicentral distance of the station

- Moment Magnitude Scale
  - More accurate for large quakes.

$$M_W = \frac{2}{3} \log M_0 - 10.7$$

http://www.fareastgizmos.com/kyocera_installs_earthquake_early_warning_system_in_all_its_japanese_locations.php

Application of Logs: Sound

- Frequency of 1000Hz is just audible when:
  Intensity $I_0 = 10^{-16}$ W/cm^2
- Noise level is given in decibel by the following:

$$L = 10\log\frac{I}{I_0}$$

- Hearing damage can occur from 85dB with long term exposure.

147.7 dB at 20 Hz 1300 feet away

- Whisper: L ≈ 20dB
- Talking: L ≈ 65 dB
  – Determine the the relationship of the intensities between talk to whisper?

$$20 = 10\log\frac{I_w}{I_0}$$

$$10^2 = \frac{I_w}{I_0}$$

$$65 = 10\log\frac{I_t}{I_0}$$

$$10^{6.5} = \frac{I_t}{I_0}$$

$$\frac{I_t}{I_w} = 10^{4.5} = 3.16\times10^4$$

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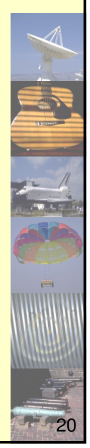
Problems to solve:

- Re-write these using the correct SI unit prefixes and 3 significant figures:
  – 54 987 299 J
  – 123 456 789 W
  – 987 654 321 999 775 km
  – 0.000 000 987 32 s
  – 0.006 237 m
  – 0.005 435 µm
- Using your own calculator determine the following:
  – $10^6 \times 10^4$
  – $10^6 / 10^4$
  – $10^6 \times 10^{-4}$
  – $6^6 \times 6^4$
  – $9^{2.5}$
  – $\sqrt[3]{8^{-2}}$
  – $8^{-2/3}$
- Using your own calculator determine the following:
  – $e^{-1.25}$
  – $e^{-0.25}$
  – $e^{0.25}$
  – $e^{1.25}$
  – $e^{2.25}$
- What trend do you observe?



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Problems to solve:

- Using your own calculator determine the following:
  - Log (549 × 23)
  - Log (549/10)
  - Log (549)/Log(10)
  - Log $(549/10)^3$
  - Log 0
  - Log (-549)
- Do you observe any trends?
  - ln (549 × 23)
  - ln (549/10)
  - ln (549)/ln(10)
  - ln $(549/10)^3$
  - ln 0
- Do you observe any similarities/differences between the Log and ln?

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Examples

- Write the following using the correct prefixes and 2 or 3 significant figures:

– 1 234 567 mm	– 1.2 km
– 46 351 259 378 bytes	– 46 Gbytes
– 0.000 003 89 m	– 3.9 μm
– 0.000 000 000 000 054 s	– 0.054 ps
– 7 891 N	– 7.9 kN
– 654 321 J	– 0.65 MJ

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Examples

- Determine the following in exponent notation:
  - $10^2 \times 10^3 = \quad 10^{2+3} = 10^5$
  - $\left(3\times10^2\right)\left(2\times10^0\right) = 3\times2\times10^{2+0} = 6\times10^2$
  - $\dfrac{a^5}{a^3} = \quad a^5 \times a^{-3} = a^{5-3} = a^2$
  - $\dfrac{10^0}{10^{-3}} = 10^0 \times 10^{-(-3)} = 10^{0-(-3)} = 10^3$
  - $\dfrac{8\times10^2}{2\times10^{-6}} = \left(8/2\right)\times10^{2-(-6)} = 4\times10^8$

Rules for Logs

- If *a, P, Q* are positive, then
  - $\log_a P^n = n\log_a P$
  - $\log_a \sqrt[n]{P} = {}^1\!/_n \log_a P$
  - $\log_a 1 = 0$
  - $\log_a \left({}^1\!/_P\right) = -\log_a P$
  - $\log_a \left(P\times Q\right) = \log_a P + \log_a Q$
  - $\log_a \left({}^P\!/_Q\right) = \log_a P - \log_a Q$
  - $\log_a P = \left(\log_b P\right)\times\left(\log_a b\right)$

Examples

- Determine the following
  - $10^3 = 1000$
  - $\text{Log } 1000 = 3$
  - $10^2 = 100$
  - $\text{Log } 100 = 2$
  - $10^{1.5432} = 34.930$
  - $\text{Log } 34.93 = 1.5432$
  - $10^0 = 1$
  - $\text{Log } 1 = 0$
  - $10^{-1.5432} = 0.02863 = 1/10^{1.5432}$
  - $\text{Log } 0.02863 = -1.5432$

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Examples

- Determine n in the following:
  - $n = \log 123.4 = 2.0913$
  - $n = \log 12.34 = 1.0913$
  - $n = \log 1.234 = 0.0913$
  - $n = \log (123.4)^2 = 4.1826 = 2(2.0913)$
  - $n = (\log 123.4)^2 = 4.3736$
  - $n = \ln 123.4 = 4.8154$
  - $n = \ln \left(1/e^2 \right) = -2$

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Problems to solve:

- Re-write these using the correct SI unit prefixes and 3 significant figures:
  - 54 987 299 J
  - 123 456 789 W
  - 987 654 321 999 775 km
  - 0.000 000 987 32 s
  - 0.006 237 m
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- Using your own calculator determine the following:
  - $10^6 \times 10^4$
  - $10^6 / 10^4$
  - $10^6 \times 10^{-4}$
  - $6^6 \times 6^4$
  - $9^{2.5}$
  - $\sqrt[3]{8^{-2}}$
  - $8^{-2/3}$
- Using your own calculator determine the following:
  - $e^{-1.25}$
  - $e^{-0.25}$
  - $e^{0.25}$
  - $e^{1.25}$
  - $e^{2.25}$
- What trend do you observe?

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Problems to solve:

- Re-write these using the correct SI unit prefixes and 3 significant figures:
  - 54.9 MJ
  - 123 MW
  - 987×10^3 Tm (987 Pm (Peta m)
  - 0.987 µs
  - 6.237 mm
  - 5. 435 nm
- Determine the following:
  - 10^{10}
  - 10^2
  - 10^2
  - $6^{10} = 6.047 \times 10^7$
  - $9^{2.5} = 243$
  - $\sqrt[3]{8^{-2}} = 0.25$
  - $8^{-2/3} = 0.25$
- Using your own calculator determine the following:
  - $e^{-1.25} = 0.286$
  - $e^{-0.25} = 0.778$
  - $e^{0.25} = 1.28$
  - $e^{1.25} = 3.49$
  - $e^{2.25} = 9.49$
- Do you observe any trends?

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Problems to solve:

- Using your own calculator determine the following:
  - Log (549 × 23)
  - Log (549/10)
  - Log (549)/Log(10)
  - Log $(549/10)^3$
  - Log 0
  - Log (-549)
- Do you observe any trends?
  - ln (549 × 23)
  - ln (549/10)
  - ln (549)/ln(10)
  - ln $(549/10)^3$
  - ln 0
- Do you observe any similarities/differences between the Log and ln?

-
  - 4.101
  - 1.74
  - 2.74
  - 5.22
  - Error / underfined
  - Error / underfined
-
  - 9.44
  - 4.01
  - 2.74
  - 12.02
  - Error / underfined

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Angles and Trig Functions

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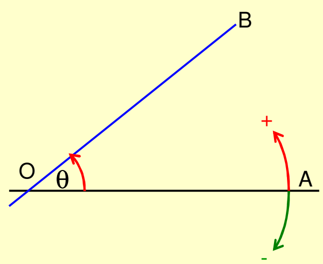
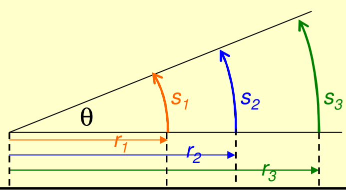
Angles



- Angle θ is positive for anti-clockwise rotation
- Angle θ is negative for clockwise rotation
- 1 Rotation = 360°
- ½ Rotation = 180°
- ¼ Rotation = 90°



$$\frac{s_1}{r_1} = \frac{s_2}{r_2} = \frac{s_3}{r_3}$$

$$\Rightarrow \frac{s}{r} = \theta$$

- Radians (rad)

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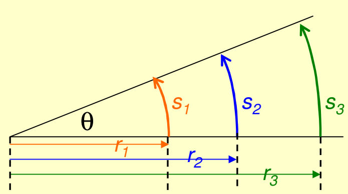
Angles - 2



$$\frac{s_1}{r_1} = \frac{s_2}{r_2} = \frac{s_3}{r_3} = \text{constant}$$

$$\Rightarrow \frac{s}{r} = \theta$$

- 1 Rotation = $\frac{\text{circ of circle}}{r} = \frac{2\pi r}{r}$

∴

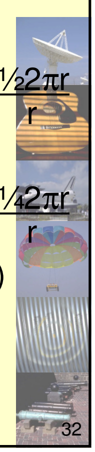
- 1 Rotation = 2π radians (360°)
- ½ Rotation = ½$\frac{\text{circ of circle}}{r} = \frac{½ 2\pi r}{r}$

  = π radians (180°)
- ¼ Rotation = ¼$\frac{\text{circ of circle}}{r} = \frac{¼ 2\pi r}{r}$

  = π/2 radians (90°)

∴

- 1° = 0.01745 rad
- 1rad = 57.3°



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Examples

- Convert to Radians
  - 1° =
  - 32° =
  - 175° =
  - 270° =
  - 315° =
- A circle has a radius of 3 cm what is the arc length subtended by an angle of 50°

- Convert to degrees
  - ½ rad =
  - ½ π rad =
  - $^4/_3\ \pi$ rad =
  - ⅔ π rad =
  - 7 rad =
  - 6 rad =
  - 1 rad =

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Calculator Skills

- Change angle setting between degrees – radian – gradient: DGR
- Convert angle between degrees – radian – gradient: DGR ►
- Set decimal and exponents: FSE
  - Fixed – Scientific – Engineering
  - Decimal – 10^{x} – 10^{3n}
- Using 10^{x}
  - 3×10^2 press 3 then *Exp* then 2
  - *Exp* $\equiv \times 10$

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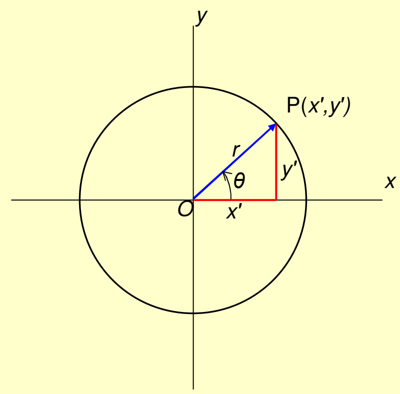
Trigonometry

- Angle can be in ° or rad.
- Make sure calculator is in appropriate mode.

$$\sin\theta = \frac{y'}{r} \qquad \cos\theta = \frac{x'}{r}$$

$$\tan\theta = \frac{y'}{x'}$$



- Make sure you know how your calculator requires the input.
  - As you see it written - Function then angle (DAL)
  - Or Angle then Function

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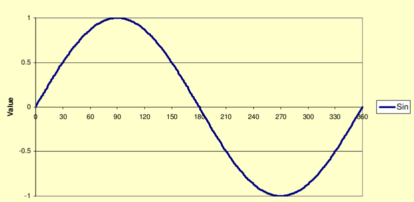
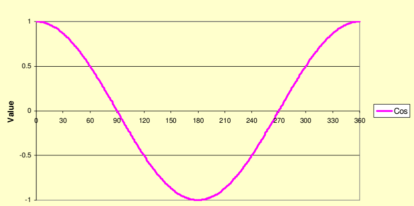
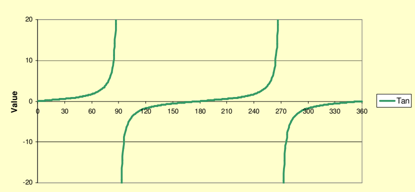
Trigonometry







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Trigonometry

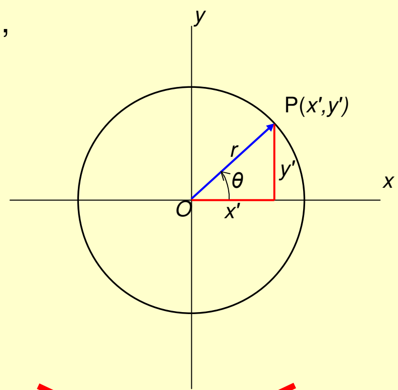
- If the relationship between the triangle sides are known, then the angles can be determined.
- The arc function:

$$Sin\,\theta = \frac{y'}{r} \qquad Cos\,\theta = \frac{x'}{r}$$

$$Sin^{-1}\left(\frac{y'}{r}\right) = \theta \qquad Cos^{-1}\left(\frac{x'}{r}\right) = \theta$$

$$Tan\,\theta = \frac{y'}{x'}$$

$$Tan^{-1}\left(\frac{y'}{x'}\right) = \theta$$



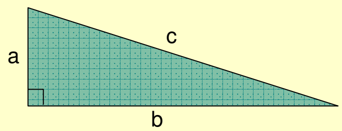
$$\cancel{Sin^{-1} a = (Sin\, a)^{-1}}$$

Examples

- Give the following answers for θ in °:
  - Sin^{-1} 0.8660 =
  - Tan^{-1} 1.000 =
  - Cos^{-1} (-0.5) =
  - Cos^{-1} 0.5 =
  - Sin^{-1} (-1.000) =
  - Tan^{-1} 0 =
  - Cos^{-1} 1.34 =
- Give the following answers for θ in radians (rad). Confirm your answer by converting to ° and compare with previous answer.
  - Sin^{-1} 0.8660 =
  - Tan^{-1} 1.000 =
  - Cos^{-1} (-0.5) =
  - Cos^{-1} 0.5 =
  - Sin^{-1} (-1.000) =
  - Tan^{-1} 0 =

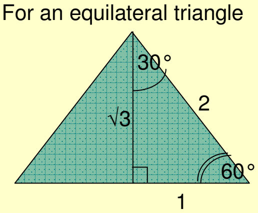
Trig and Triangles

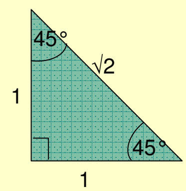
For a right angled triangle



Pythagoras Theorem:
$c^2 = a^2 + b^2$

For an equilateral triangle





Example

- Complete the following table using the special triangles on previous page.

θ °	0	30	45	60	90	180
θ rad	0	π/6	π/4	π/3	π2	π
Sinθ						
Cos θ						
Tan θ						

Trig. Identities

- Not used very often in this course – but good to know them when needed.

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

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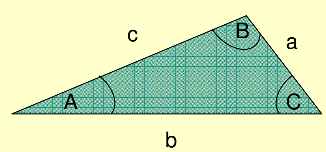
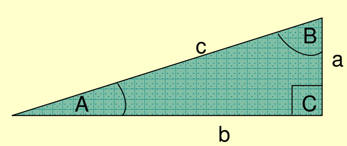
Trig. Identities

- Cosine Rule
  - $c^2=a^2 + b^2 -2ab\cos C$

  If C = 90°
  - ????



- Sine Rule
  - $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$



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Examples

- If $A = B$ derive, using trig identities, an equation for:
  - Sin $2A$
  - Cos $2A$
- Write the following in terms of cos $2A$
  - $\sin^2 A =$
  - $\cos^2 A =$

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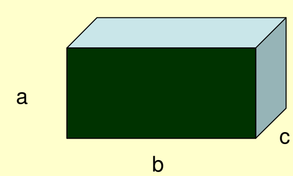
Geometric Formulae

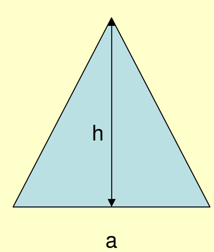
- Rectangular shapes
  - Area = ab
  - *Volume = Area × depth*

    $= abc$



- Triangular shapes
  - Area = ½base × *height*

    $= \frac{1}{2}ah$



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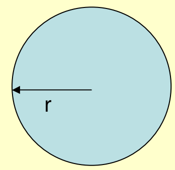
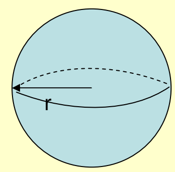
Geometric Formulae-2

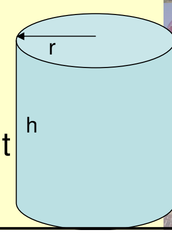
- Circle
  - Circumference $=2\pi r$
  - Area $= \pi r^2$



- Sphere
  - Area $= 4\pi r^2$
  - Volume $= {}^4/_3\pi r^3$

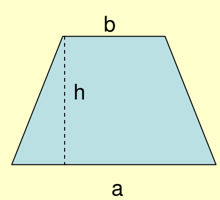


- Cylinder
  - Area $= 2(\pi r^2) + 2\pi rh$
  - Volume = cross-sectional area × height
  - Volume $= \pi r^2 h$



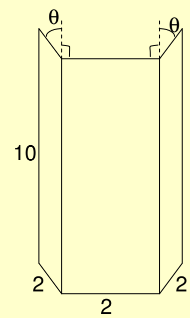
Geometric Formulae-3

- Trapesium
  - Area $= ½(a+b)h$



Examples

- A flat metal plate (dimensions in cm) is bent in the form of a gutter (see figure below). Find a **formula** for the volume of the gutter in terms of the angle θ.



Examples - 2

- Consider an equilateral triangle with side lengths of 2 m
  - *Using the angles of the triangle, determine the area of the triangle.*
  - *Without using the angles of the triangle, determine its area.*

Examples – 3

- A circular racetrack has an inner radius of 2 km and track width of 6 m. Determine the area of the roadway in square meters.

Examples – 4

- An empty cylindrical pipe has an outer diameter of 30 mm and an inner diameter of 25 mm.
  - *Determine the volume of material needed to produce a 1 m length of this pipe.*
  - *Determine the area of the pipe exposed to the atmosphere.*
  - *What is the volume of water which 2 m of this pipe would contain if it were closed at one end?*

Examples

- Convert to Radians
  - 1° = 0.01745 rad
  - 32° = 0.5585 rad
  - 175° = 3.054 rad
  - 270° = 4.7124 rad
  - 315° = 5.4977 rad
- A circle has a radius of 3 cm what is the arc length subtended by an angle of 50°

$$\theta = 50° = 0.8727\,\text{rad}$$

$$= \frac{s}{r}$$

$$s = \theta r = (0.8527)(3\,\text{cm})$$

$$= 2.6\,\text{cm}$$

- Convert to degrees
  - ½ rad = 28.65°
  - ½ π rad = 90°
  - $^4/_3$ π rad = 240°
  - ⅔ π rad = 120°
  - 7 rad = 401.1°
  - 6 rad = 343.8°
  - 1 rad = 57.3°

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Examples

- Give the following answers for θ in °:
  - Sin^{-1} 0.8660 = 60°
  - Tan^{-1} 1.000 = 45°
  - Cos^{-1} (-0.5) = 120°
  - Cos^{-1} 0.5 = 60°
  - Sin^{-1} (-1.000) = -90°
  - Tan^{-1} 0 = 0°
  - Cos^{-1} 1.34 = Undefined/Error
- Give the following answers for θ in radians (rad). Confirm your answer by converting to ° and compare with previous answer.
  - Sin^{-1} 0.8660 = 1.047 rad
  - Tan^{-1} 1.000 = 0.7855 rad
  - Cos^{-1} (-0.5) = 2.094 rad
  - Cos^{-1} 0.5 = 1.047 rad
  - Sin^{-1} (-1.000) = 1.571 rad
  - Tan^{-1} 0 = 0 rad

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Example

- Complete the following table using the special triangles on previous page.

θ °	0	30	45	60	90	180
θ rad	0	π/6	π/4	π/3	π2	π
Sinθ	0	½	$\sqrt{2}/2$	$\sqrt{3}/2$	1	0
Cos θ	1	$\sqrt{3}/2$	$\sqrt{2}/2$	½	0	1
Tan θ	0	$1/\sqrt{3}$	1	$\sqrt{3}/1$	Undef.	0

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Examples

- A flat metal plate (dimensions in cm) is bent in the form of a gutter (see figure below). Find a **formula** for the volume of the gutter in terms of the angle *θ.*

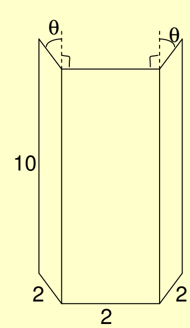
$$Sin\theta = \frac{base}{2}$$
$$base = 2\sin\theta$$

$$\cos\theta = \frac{height}{2}$$
$$height = 2\cos\theta$$

$$Volume_{side\Delta} = ½ base \times height \times length$$
$$= ½(2\sin\theta)\times(2\cos\theta)\times 10$$
$$= 20\sin\theta\cos\theta$$

$$Volume_{centre} = base \times height \times length$$
$$= 2\times 2\cos\theta \times 10$$
$$= 40\cos\theta$$

$$Volume_{total} = 2\times 20\sin\theta\cos\theta + 40\cos\theta$$
$$= 40\cos\theta(\sin\theta + 1)$$



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Examples - 2

- Consider an equilateral triangle with side lengths of 2 m
  - *Using the angles of the triangle, determine the area of the triangle.*
  - *Without using the angles of the triangle, determine its area.*

Angles in equilateral triangle: all 60°
Height of triangle determined from
Sin θ = h/2
h = 2 sin 60°
Area = ½ base x h

Use Pythagoras theorem to determine height
Area = ½ base x h

Ans: 1.732 m^2

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Examples – 3

- A circular racetrack has an inner radius of 2 km and track width of 6 m. Determine the area of the roadway in square meters.

Decide which units you want to use.
Determine inner area of track.
Determine outer area of track.
Find difference between two areas.
Ensure that your final answer is in meters

$$Area_{inner} = \pi r_{in}^2$$

$$Area_{outer} = \pi r_{out}^2$$

$$Area_{road} = \pi\left(r_{out}^2 - r_{in}^2\right)$$

Ans: 75 500 m^2

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Examples – 4

- An empty cylindrical pipe has an outer diameter of 30 mm and an inner diameter of 25 mm.
  - *Determine the volume of material needed to produce a 1 m length of this pipe.*
  - *Determine the area of the pipe exposed to the atmosphere.*
  - *What is the volume of water which 2 m of this pipe would contain if it were closed at one end?*

Ans: 2.75×10^{-4} m^3
0.189 m^2
3.92×10^{-3} m^3 water

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